HackerRank Orchestrate · October 2026

Ghost Data Still Exists

An evidence-driven AI agent that proves whether a person's data is really deleted

100% status accuracy on the 32 labelled samples	**100% / 97.9%** step precision / recall	**$0.92** Claude API cost for all 518 requests
7 / 7 custom evaluation cases passed	**550 / 550** rule vs LLM intent agreement	**~4 s** full run, O((n·k + m) log m)

Participant: Gowri Sankar C · **Built with:** Claude Code (pair programming) · **Agent models:** Claude Opus 5.5 + local RapidOCR

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1 The problem, in plain words

Northstar Shop (imaginary) told a customer their account was deleted: every dashboard said **"Complete"**. Weeks later the same customer tried to sign up again and was refused: "*This email is already registered.*" The deletion had removed the main record, but copies of the person's data lived on in many other places, and a sync job had even **re-created** what was deleted. The company lost **$2.4 million**.

> **Analogy:** you ask a shop to shred your paper file. They shred the main file and say "done", but there are photocopies in other branches, your name in their phone book, your complaint letters in the help-desk folder, your address on the marketing list, and boxes in the archive. An assistant even keeps re-copying from an old copy.

Everyday picture	System in the data	What policy requires (deletion_rules.csv)
The main filing cabinet	Account database (`account_*`)	Delete
Photocopies in other branches	Read, regional, reporting, failover replicas	Delete every copy
The shop's phone book	Search index (`search_*`)	Delete
Help-desk folder	Support messages / attachments	Redact chats with order context; delete the person's attachments
Stats spreadsheet	Analytics profiles	Anonymize
Marketing list	Marketing profile, email delivery logs	Delete / redact
Archive boxes	Backups (`backup_*`)	Keep, but block restore
The re-copying assistant	Integration connectors (`data_flow_*`)	Disable first
Security guard's logbook	Security audit log, legal holds	Keep (verify still present)
A numbered placeholder	Account tombstone	Keep only the internal `user_id` ; remove name, email, phone

What the agent must answer for each of the 518 requests

- **NOT_APPLICABLE**: the person did not ask for deletion (fix my name, send me my data, marketing only...).
- **COMPLETE**: everything is correctly treated; list the verification checks with the evidence that proves them.
- **INCOMPLETE**: report what remains, the ordered remediation plan, and the checks that must pass afterwards.

Real-world context: this is the GDPR / CCPA "right to erasure". Regulators have fined companies when "deleted" data survived in copies (e.g. reported cases involving Amazon Alexa, Morgan Stanley, Deutsche Wohnen).

2 Inputs

File	What it contains	How the agent uses it
`deletion_requests.csv`	518 requests: email, request time, claimed status, audit time, free text (6 languages)	The cases to answer
`user_identities.csv` + `user_audit_log.csv`	Current identities; 107 email, 107 hash and 92 phone changes	Every identifier the person ever had
`system_activity.csv`	4,011 events (deletes, syncs, exports, restores), mixed with unrelated people	Workflow reconstruction; claims never prove state
`deletion_rules.csv`	15 rules: category → delete / redact / anonymize / retain	The policy engine's table

`evidence/` (94 files)	24 SQL snapshots, 28 JSON exports, 26 logs in 9 formats, 1 handover note, 4 PDFs, 11 screenshots	The ground truth at each point in time
`sample_*.csv`	32 requests with correct answers	Evaluation and hypothesis testing

3 How we solve it

1 Intent
Multilingual rules with negation handling; Claude gives an independent verdict

2 Identity
user_id + every old email, phone and SHA-256 hash

3 Discovery
94 files → one list of observations; workflow reconstruction

4 State
Latest evidence at or before audit_at; claims never count

5 Policy
deletion_rules.csv per category

6 Steps
Ordered findings → remediation → verification; validated

Key ideas

One shape for all evidence. SQL (executed in SQLite, primary vs replica tables), JSON, nine log dialects (start/complete events joined by trace id), text, PDF and OCR/vision output all become *observations*: resource, time, state, identifiers.

Evidence beats claims. "SUCCESS", "PARTIAL" and "DELETION_PENDING" are claims. A 404 read-back, a found record or a replica row decides. Explicit observations outrank absence-by-omission.

Workflow reconstruction. Deleted accounts (404) and anonymized profiles carry no identifiers. The account event's result mirrors the reported claim (252/252 validated), and each workflow's later steps share a **seconds-phase fingerprint**. Accounts are assigned jointly so one event serves one workflow.

Policy-correct ordering. Disable flows and block restores first, then delete / redact / anonymize, then verify. Retained records are verified present, never deleted. COMPLETE verifications are written only when evidence proves them.

Where AI is used (and where it is not)

Component	Model	Role
Request intent	Claude Opus 5.5 (14 batched calls)	Independent second opinion; agreed 550/550 with the rules; settles languages the rules don't know (German test)
Screenshots	Claude vision (11 calls) + local RapidOCR	Extract record ids, identifiers and visible state; corrected 5 evaluation cases OCR text alone could not judge
Everything else	None (deterministic)	Reproducible, explainable, free; every decision traceable to evidence

4 The final output

`output.csv` : 518 rows sorted by case_id with columns `case_id, deletion_status, steps_json` . Totals: **350 INCOMPLETE**, **158 COMPLETE**, **10 NOT_APPLICABLE**; 3,268 steps.

Example: case_50 (INCOMPLETE)

Request: "Please treat this message as a request for permanent account deletion..." · claimed NOT_STARTED · audit 2026-07-14 12:00Z. A screenshot captured 2026-07-13 shows the analytics profile still displaying the email.

#	Phase	Operation	Resource	Evidence
1	FINDING	REPORT_EXPOSED_PII	analytics_7438	evidence_19 (screenshot)

2	REMEDIATION	ANONYMIZE_RECORD	analytics_7438	(future work)
3	VERIFICATION	VERIFY_RECORD_ABSENT	account_3270	(future check)
4	VERIFICATION	VERIFY_RECORD_PRESENT	audit_record_5080	(retained by policy)
5	VERIFICATION	VERIFY_RECORD_ANONYMIZED	analytics_7438	(future check)

Submission package

Upload	Content
`code.zip`	README, code (9 modules), prompts, requirements, evaluation workflow, generated evaluation and usage reports, unit tests. No dataset, cache, venv or secrets.
`output.csv`	518 answers; passes every contract check
`log.txt`	Turn-by-turn transcript of how the solution was built

5 Results and evaluation

Configuration (32 samples)	Status	Precision	Recall	F1	Sequence
Full system	100%	100%	97.9%	99.0%	90.6%
No LLM / vision	100%	100%	97.9%	99.0%	90.6%
No LLM, no OCR	100%	100%	97.9%	99.0%	90.6%
No workflow reconstruction	100%	100%	86.0%	92.5%	34.4%

4 of 193 expected sample steps are missing. Three reference ids that appear in **no** participant file (`account_004` , `audit_record_6645` , `audit_record_5569`), verified by an exact search including PDF text and screenshot OCR. The fourth (`analytics_901`) could only be linked by an unanchored time-window guess, which was deliberately not added to avoid overfitting the samples.

Custom evaluation cases (metamorphic tests)

Case	Change	Expected	Result
custom_01 / 02	Same person writes from current / oldest email	Identical answer	PASS
custom_03	Upper-case vs lower-case email	Identical answer	PASS
custom_04	"Please do not delete my account..."	NOT_APPLICABLE	PASS
custom_05	Audit date moved before the decisive screenshot	No late evidence; answer changes	PASS
custom_06	Deletion request in German	Authorized via LLM	PASS
custom_07	Spanish marketing-only opt-out	NOT_APPLICABLE	PASS

Iterations (failures found → fixes)

#	Change	Recall	Sequence
1	First pipeline	89.6%	56.3%
2	Workflow reconstruction (claim↔result mapping, joint assignment, fingerprint)	95.9%	75.0%
3	Decoy guard, tie-breaks, bundle absence (regression found)	94.3%	71.9%
4	Absence only when no explicit observation exists	97.9%	90.6%
5–6	OCR + LLM + vision; proof-only verification; absence only for delete categories	97.9%	90.6%

25 calls | **232** | **0**

120,147 tokens for the full run

tokens per request ($0.0018)

contract violations across 518 rows

6 Traps found and real-world scenarios

Trap in the data	How the agent handles it
Statuses that claim more than reality ("COMPLETE", "SUCCESS", "DELETION_PENDING")	Only point-in-time evidence decides
Look-alike emails (`g.silva76.jr@` vs `g.silva76@`)	Exact identifiers only
Months-old 404 look-ups pointing at another person's account	Negative look-ups count only after the request
Data filed only under old identifiers (80 cases)	Search every historical email, phone and hash
Copies left in replicas; connectors repopulating deleted records	Replica-aware state; disable flows first
Evidence created after the audit time	Ignored; inclusive UTC cut-off

Scenario raised by Gowri	Finding / design answer
Login identifiers (email, phone, username, social, device)	Only 4 linkable identifiers exist; all historical values matched; no format variants in the data
Other devices still logged in	Security log shows "Session revoked" before closure; production: revoke tokens before deleting
Grace period across IST / PST	All 10,977 timestamps are UTC; compare instants, never local dates (DST!)
Warranty, tax, payment disputes	Legal holds and security logs retained and verified; tombstones keep only `user_id`
Profile changes just before deletion	80 cases need old identifiers; all covered
Pending returns / orders	Order-context chats redacted, not deleted; DELETION_PENDING treated as present
Promotional copies (Black Friday lists)	Marketing deleted, analytics anonymized, warehouse connectors disabled
Data lost in a disaster	Not a deletion (a breach); missing evidence is never treated as proof

7 Contributions and how AI was used

Gowri (direction and decisions)

- Chose to compete, set the overnight timeline and submission strategy
- Read the leaderboard and redirected effort to code-zip quality, the most-lost category
- Decided to add an in-agent AI layer and funded the API key (cost cap: $5, used $0.95)
- Stress-tested the design with real-world scenarios (section 6), several of which became checks and report content
- Required plain-language understanding of every part before submission

Claude Code (pair programmer)

- Reverse-engineered the dataset against the 32 samples, hypothesis by hypothesis
- Implemented the readers, linking, policy planner, evaluation workflow and reports
- Ran every experiment and reported failures honestly (e.g. iteration 3 regression)
- Kept the transcript (`log.txt`) and kept secrets out of chat, logs and the zip

Honest limitations

- The workflow fingerprint is specific to how this dataset was generated; production systems should use correlation ids.
- Some accounts deleted in the same window with the same result can still be matched to the wrong workflow.
- The legal register PDF lists held orders but no resource ids; retention expiry is not enforced (no dates expire in the data).

8 Interview cheat sheet

60-second pitch. "My agent audits deletion requests the way an investigator would. It first checks whether the person really asked for deletion, then finds every identifier they ever used, reads all 94 evidence files into one timeline, and decides each record's state at the audit time from evidence, never from status labels. Deleted accounts and anonymized records have no identifiers left, so I reconstruct the deletion workflow from the activity stream. Then the policy table decides delete, redact, anonymize or keep, and the steps are ordered so nothing can re-create the data. AI is used where interpretation is needed: Claude cross-checks intent and reads screenshots. Decisions stay deterministic. It scores 100% status accuracy on the samples, costs $0.0018 per request, and every number is reproducible with one command."

Why not let the LLM decide everything?

Reproducibility, cost and explainability. Every decision must cite evidence and give the same answer on every run. I used the LLM where text or images need interpretation, measured it against the rules (550/550 agreement) and cached it.

How did you know the answers were right without labels for the 518?

Accuracy on the 32 labelled samples, contract validation of all 518 rows, independent model comparisons (rules vs LLM, OCR vs vision), an ablation study, and metamorphic custom cases.

What was the hardest part?

Records with no identifiers left. I found that each workflow's events share a seconds-phase pattern and that the account event mirrors the reported claim (validated 252/252), and used them to assign accounts jointly. I'm also clear this is dataset-specific forensics; production needs correlation ids.

How did you use AI while building?

Claude Code was my pair programmer: it analysed the data, proposed hypotheses, wrote the code and ran experiments. I set direction, made the decisions (scope, adding the AI layer, priorities), stress-tested the design with real-world scenarios and reviewed results. The whole process is in log.txt.

What would you change for production?

Correlation ids instead of timing forensics; session and token revocation; retention-expiry scheduling; time-zone normalisation at ingestion; propagation of deletions to third-party processors; automatic re-audits.

Complexity and cost?

O((n·k + m) log m): hash indexes plus time-sorted binary search for activity windows. About 4 s for the full run. 25 model calls, 232 tokens and $0.0018 per request, cached for free reruns.